

(10)

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Deep Learning

C02-Cloggs Test - AT₃

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Machine Learning

Deep Learning

1. It uses shallow models such as Decision Trees, SVM, Random Forest.

1. uses deep Neural networks with multiple hidden layers.

2. Perform well with small to medium datasets.

2. Perform well with small to large datasets.

3. Features are manually selected by domain experts.

3. Features are automatically learned

4. Faster training

4. Longer training due to multiple layers.

5. CPU is usually sufficient

5. GPU is Generally Required

Ex: 1. Email spam detection

Ex:

2. Credit card fraud detection

1. Self driving cars

2. Face recognition system

2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.

A. Representation Learning is the process of neural network automatically learns useful features from raw input data instead of relying on manually designed features.

Ex:

Instead of manually identifying edges, textures and shapes in images, a CNN automatically learns these features in training.

Advantages:

1. ~~Eliminates~~ Eliminates manual feature engineering.
2. Learning hierarchical features.

Effects of width and depth:

Width:

It refers to the number of neurons in each hidden layer.

Influence of width

1. Learning capability (Learning more features at one level)
2. Computational complexity (Require more memory)
3. Model performance (Improves accuracy initially)

Depth:

Depth refers to the number of hidden layers.

Influence of depth

1. Learning hierarchical and abstract features
2. Longer training time
3. Usually provides higher accuracy

3. Compare Relu, Leaky Relu (LRelu) and Elu activation functions

Explain their mathematical behaviour, advantages, disadvantages and suitable use cases.

A) Relu:

$$f(x) = \max(0, x)$$

Leaky Relu:

$$f(x) = x, x > 0; \alpha x, x < 0$$

Elu:

$$f(x) = x, x > 0; \alpha(e^x - 1), x < 0$$

Relu:

Behaviour

Positive values remain unchanged.

Negative values become zero.

Advantages

- Faster computation
- Reduces vanishing gradient.
- Sparse activation.

Disadvantages

Dead neuron problem.

Application

CNN

Image Classification

Leaky Relu

Behaviour

Positive values unchanged

Negative values retain in small slope

Advantages

Prevents dead neurons

Disadvantages

Requires choosing α

Application

Deep CNNs

Elu

Behaviour

1) Positive inputs remain ~~linear~~ linear

2) Negative outputs become smooth exponential values.

Advantages

Computationally more expensive

Applications

Deep neural networks.

4. Explain the concept of unsupervised learning in neural networks.
Describe the working principle of Restricted Boltzmann machines and Autoencoders and compare their objectives and applications.

A) Unsupervised Learning

It is a learning method where the neural network is trained using unlabeled data.

Applications:

1. Customer segmentation
2. Data compression.
3. Anomaly Detection.

Restricted Boltzmann machines :-

An RBM is a generative stochastic neural network used to learn the probability of distribution of input data.

Structure

1. Visible Layer
2. Hidden Layer
3. Reconstruction Layer, error

Applications

Recommendation systems

Feature extraction

Dimensionality reduction.

Autoencoder

It is an unsupervised neural network that learns to reconstruct its input through a compressed representation structure

Encoder

Bottleneck

Decoder

Working principle

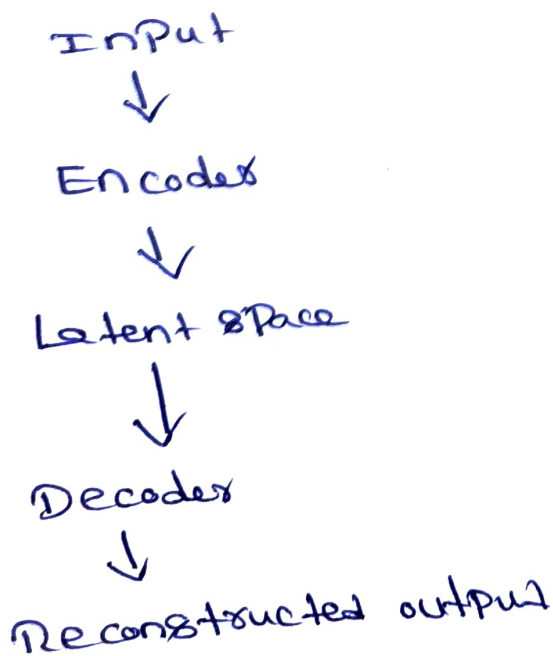
1. Encoder compresses input
2. Latent space stores essential information
3. Decoder reconstructs the original input
4. Reconstruction error is minimized.

5. Explain the concept of Autoencoder and its uses in Deep Learning

Autoencoder

It is an supervised neural network that learns to reconstruct its input through a compressed representation of input data and Reconstructed to original input with minimum reconstruction error

Architecture



Working

1. Input data is Given to encoder
2. The encoder compresses data
3. The decoder reconstructs the original data from latent representation
4. The network minimizes the reconstruction loss.

Advantages

1. Automatic feature Learning
2. Reduces dimensionality
3. Removes noise from data

Application

1. Image denoising
2. Image compression
3. Dimensionality Reduction